

Project Proposal

Title: *FABLE (Flexible AI-Based Language Explorer): A Web Text Adventure Game Powered by NLP*

Course: COMP5423 Natural Language Processing

Date: January 2026

1. Executive Summary

This project proposes **FABLE**, a web-based, AI-driven text adventure game that allows players to interact with a dynamic narrative using natural language. FABLE combines **intent understanding, entity recognition, dialogue management, and neural text generation** to produce coherent, branching stories that adapt to player actions. Inspired by the design patterns of modern AI text-adventure platforms like **Fiction World AI**—which emphasize **AI-powered narratives, voice input, text-to-speech, private adventures, and editable AI responses**—FABLE will demonstrate the practical application of NLP techniques in an engaging, playable system that runs in the browser and a lightweight backend.

FABLE's research components include: (1) an **NLP pipeline** (intent classification + NER) for semantic control, (2) a **Dialogue Manager** for maintaining game state and plot consistency, (3) a **Story Generator** (LLM-based) with guardrails for safety and coherence, and (4) **evaluation protocols** for narrative quality and user experience.

2. Motivation and Background

Text adventure games are a natural fit for NLP: players express intentions in free-form language, while the system must interpret, ground, and produce narrative responses. Recent AI story platforms (e.g., **Fiction World AI**) highlight player-driven stories, **voice interactions** (speech-to-text for input, text-to-speech for narration), **creative control** (edit/regenerate model output), and **community-authored scenarios**—all of which point to a compelling pedagogical setting for an NLP course project and a strong baseline for FABLE's feature set.

3. Objectives

3.1 Primary Objectives

- Playable Web Game:** Deliver a browser-based text adventure with natural-language input and AI-generated narrative output.
- NLP Pipeline:** Build an interpretable pipeline including **Intent Classification, Named Entity Recognition (NER), and Dialogue/State Management**.
- Story Generation:** Integrate a controllable **LLM-based generator** to produce coherent narrative passages conditioned on game state and player intent.

4. **Multimodal UX:** Provide **speech input** and **text-to-speech (TTS)** narration for accessibility and immersion, following the capability patterns used by platforms like Fiction World AI.
5. **Evaluation:** Design and run qualitative and quantitative evaluations (e.g., coherence, responsiveness, user satisfaction).

3.2 Secondary Objectives (Stretch)

- **Scenario Builder:** A minimal creator tool to define custom worlds, items, NPCs, and victory conditions (in JSON).
 - **Content Guardrails:** Toxicity/safety filtering and style control.
 - **Editable AI Output:** Allow users to “Edit” or “Regenerate” an AI turn, mirroring modern platform affordances.
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4. Scope

- **In Scope:** Single-player adventures; English gameplay text; browser UI; small scenario library; local account system (or session-based saves); lightweight backend APIs; small-scale model experimentation.
 - **Out of Scope:** MMO-scale features; complex graphics; high-cost proprietary APIs (unless academic credits available); large social features; mobile native packaging (PWA acceptable).
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5. Gameplay and User Experience

5.1 Core Loop

1. **Opening Scenario:** The player selects a scenario (e.g., *Mistwood Forest*).
2. **Natural-Language Turn:** The player types or speaks an action (“light the torch and push deeper into the fog”).
3. **NLP Pipeline:** Intent + entities are extracted and fed into the Dialogue Manager.
4. **Story Generation:** The LLM generates a narrative paragraph constrained by state and scenario rules.
5. **Present & Control:** The interface displays the story, offers **TTS playback**, and optional buttons: **Regenerate** or **Edit AI Text**.
6. **Repeat** until win/lose/end conditions are met.

5.2 UX Elements

- **Story Pane** (scrolling), **Input Box**, **Mic Button** (voice input), **Play Icon** (TTS), **Regenerate**, **Edit**, **Save/Load**, **Inventory & Stats**, **Mini-Map (optional)**.
- **Accessibility:** font scaling, dark mode, keyboard shortcuts; TTS for low-vision users.

6. System Architecture

6.1 High-Level Diagram (textual)

[Web Client (React/Vue)]

└─ Input UI (Text + Voice) —> [API Gateway]
└─ Output UI (Story + TTS) ↳ [Auth/Session]
└─ Local Save (optional)

[Backend (FastAPI/Express)]

└─ NLP Engine
| └─ Intent Classifier (lightweight classifier)
| └─ NER (spaCy / transformer NER)
| └─ Normalizer (coref-lite / pronoun handling)
└─ Dialogue Manager (State + Rules + Memory)
└─ Story Generator (LLM + prompts + constraints)
└─ Safety/Style Filter
└─ Scenario Repository (JSON)
└─ Persistence (PostgreSQL/Supabase or file store)

6.2 Technology Choices

- **Frontend:** React (Vite) or Next.js; Web Speech API for STT/TTS; Tailwind or Bootstrap.
 - **Backend:** Python **FastAPI** (or Node **Express**).
 - **NLP:** spaCy / Hugging Face Transformers; light RNN/Transformer or a small LLM via API (configurable).
 - **Storage:** PostgreSQL (via Supabase) or SQLite for simplicity.
 - **Deployment:** Docker; single-node deployment on a student server or cloud free tier.
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7. NLP & Story Generation Design

7.1 Input Understanding

- **Intent Classes (initial):** MOVE, OBSERVE, INTERACT, TALK, COMBAT, ITEM_USE, INVENTORY, META (help/save).
- **NER Targets:** PERSON/NPC, LOCATION, ITEM, CREATURE, EVENT.
- **Preprocessing:** lowercasing (preserve proper nouns), punctuation normalization, lightweight coreference hints, safety keyword screening.

7.2 Dialogue Manager (DM)

- Maintains **World State**: location graph, NPC dispositions, inventory, flags (e.g., has_torch = true).
- Applies **Scenario Rules**: success/failure conditions, resource constraints, encounter tables.
- Produces a **structured control message** for the generator:

JSON

```
{  
  "context_summary": "...",  
  "current_location": "foggy_path",  
  "visible_entities": ["wolf", "old_gate"],  
  "player_intent": "OBSERVE",  
  "salient_entities": ["torch"],  
  "style": "dark_fantasy",  
  "constraints": ["avoid gore", "PG-13 tone"]  
}
```

7.3 Story Generator

- **LLM Prompting Strategy:**
 - **System Prompt:** world tone, narrative persona, safety/style rules.
 - **Context Window:** rolling summary + last K turns.
 - **Planner + Realizer (optional):** a brief outline sentence before full narration to reduce drift.
- **Control Tokens/Tags (if supported):** tonality, pacing, POV.
- **Error Handling:** temperature bounds, max tokens, regeneration with varied seed.
- **Editable Output:** UI allows user edits (as in Fiction World AI's creative control feature).

7.4 Safety, Coherence, and Hallucination Controls

- **Safety Filter:** rule-based keyword checks + toxicity classifier (small model).
- **Coherence:** short "state consistency" checks (e.g., forbid referencing items not yet acquired).

- **Summarizer:** periodic state/narrative summaries to keep history compact and consistent.
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8. Data and Scenarios

8.1 Scenario JSON Schema (excerpt)

JSON

```
{  
  "id": "mistwood_forest",  
  "title": "Mistwood Forest",  
  "opening": "You wake in a fog-draped grove. A distant howl cuts the silence.",  
  "locations": ["grove", "foggy_path", "abandoned_temple"],  
  "items": ["torch", "map", "rusted_key"],  
  "npcs": ["mysterious_figure", "wolf"],  
  "win_conditions": ["recover_ancient_artifact"],  
  "lose_conditions": ["hp<=0", "time_limit_exceeded"],  
  "style": "dark_fantasy"  
}
```

8.2 Training / Dev Data

- We will **not** curate large proprietary datasets. Instead, we use:
 - Small handcrafted intent/NER examples for few-shot or lightweight supervised training.
 - Publicly available, permissively licensed datasets (if needed) for NER fine-tuning (e.g., CoNLL-style), limited to course scope.
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9. Evaluation Plan

9.1 Intrinsic NLP Metrics

- **Intent Classification:** accuracy/F1 on a small labeled validation set.
- **NER:** entity-level precision/recall/F1.

9.2 Extrinsic / User-Centered Metrics

- **Narrative Coherence:** human rating (Likert 1–5) on continuity and consistency.
- **Responsiveness & Control:** perceived alignment to player intent; ease of steering the story.
- **Enjoyment & Immersion:** overall satisfaction, willingness to replay.

- **Latency:** mean response time target < **3 seconds** for typical turns.

9.3 Study Setup

- **Participants:** 8–12 classmates/friends.
 - **Protocol:** each plays two short adventures (~10 minutes each), then completes questionnaires; logs collected for analysis.
 - **Analysis:** report descriptive stats; compare two model settings (e.g., small local model vs. API LLM) for quality vs. latency trade-offs.
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10. Implementation Plan & Timeline

Week	Milestone	Details
1	Project Setup	Repo, DevOps (Vite/React + FastAPI), base UI scaffold
2	NLP v0	Intent/NER prototypes; scenario JSON loader
3	Dialogue Manager v1	State store, rules, action reducer; end conditions
4	Story Generator v1	Prompt templates; safe defaults; regeneration/edit hooks
5	Voice & TTS	Web Speech API STT; TTS playback; accessibility polish
6	UX Iteration	Inventory panel, save/load, dark mode
7	Evaluation Prep	Surveys, logging, ablation toggles (model A/B)
8	User Study & Analysis	Run tests, compute metrics, finalize report & demo

Total duration: ~8 weeks (can be compressed to 4–6 with tighter scope).

11. Risks and Mitigations

Risk	Impact	Mitigation
Model drift / incoherence	Breaks narrative continuity	DM constraints; planner outline; state checks
Latency too high	Poor UX	Smaller models; caching; shorter outputs; streaming
Safety concerns	Inappropriate content	Rule-based filters + toxicity classifier; stricter prompts
Speech API variability	Recognition errors	Optional manual edit; language model spelling correction

Risk	Impact	Mitigation
Scope creep	Delays	Lock MVP: one polished scenario; advanced features optional

12. Ethical, Privacy, and Accessibility Considerations

- **Privacy:** Local sessions and minimal PII; “private adventures” ethos aligns with modern AI story platforms’ privacy framing.
 - **Content Safety:** Clear PG-13 style guardrails; user reporting; toxicity checks.
 - **Accessibility:** Font scaling, high-contrast mode, **TTS** narration; keyboard navigation.
 - **Academic Integrity:** Clearly label **AI-generated** vs. scripted content; include model cards/configs in the appendix.
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13. Expected Deliverables

1. **Playable Web App** (hosted demo or local Docker).
 2. **Source Code** (frontend + backend).
 3. **Technical Report** (NLP pipeline, prompts, evaluation results).
 4. **User Documentation** (readme, quickstart).
 5. **Short Demo Video** (2–4 minutes walkthrough).
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14. Minimal API Sketch (for clarity)

```
POST /api/turn

Request: {
  "session_id": "abc",
  "user_text": "light the torch and move forward",
  "speech": false
}

Response: {
  "story_text": "You raise the torch. Warm light pushes back the fog as you step onto the path...",
  "state_delta": {"location": "foggy_path", "torch_lit": true},
  "choices": [],
```

```
"safety": {"flagged": false}
```

```
}
```

```
POST /api/regenerate
```

```
POST /api/edit
```

```
GET /api/scenarios
```

```
POST /api/save
```

```
GET /api/load
```

15. Budget (Student Scale)

- **Hosting:** Free tier or student credits.
 - **Models:** Prefer open/local options; if using an API LLM, cap usage to a small budget ($\leq$ USD \$20).
 - **Misc:** Domain optional (USD \$10–15, not required).
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16. Success Criteria

- **Functional:** End-to-end playable adventure; stable NLP pipeline; <3s average response in test conditions.
 - **Quality:** Coherence $\geq 3.5/5$ average in user study; intent accuracy $\geq 80\%$ on validation set.
 - **Usability:** $\geq 4/5$ average satisfaction; positive qualitative feedback on control and immersion.
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17. References

- Fiction World AI overview and feature list (AI-driven narratives; voice input; TTS; community scenarios; private adventures; editable outputs):
 - Reddit announcement and features:
<https://www.reddit.com/r/FictionWorldAI/comments/1av27aw/introducing_fiction_world_ai/>
 - AlternativeTo product page (features & pricing descriptors):
<<https://alternativeto.net/software/fiction-world-ai/about/>>
 - Third-party tool description (voice input, TTS, private adventures, community scenarios):
<<https://eliteai.tools/tool/fiction-world-ai>>
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18. Appendix

A. Prompt Template (Example)

System: You are a narrative engine for a PG-13 dark-fantasy text adventure.

- Maintain continuity with the provided world state.
- Avoid gore, sexual content, and modern slang.
- Keep paragraphs under 120 words.

Context Summary: {{summary}}

Current Location: {{location}}

Visible Entities: {{entities}}

Player Intent: {{intent}}

Salient Entities: {{salient_entities}}

Style: {{style}}

Assistant: Generate a single immersive paragraph describing the outcome of the player's action.

B. Lightweight Intent Labels (Initial Few-Shots)

"move north", "go into the temple" -> MOVE

"look around", "inspect the altar" -> OBSERVE

"talk to the figure", "ask about the key" -> TALK

"use torch", "light the torch" -> ITEM_USE

"attack the wolf", "defend" -> COMBAT

"open inventory", "check bag" -> INVENTORY

"save game", "help" -> META

C. Evaluation Survey (Excerpt)

- The story reacted appropriately to my actions. (1–5)
- The narrative felt coherent and consistent. (1–5)
- I felt in control of the story's direction. (1–5)
- The system's responses were timely. (1–5)
- Overall enjoyment. (1–5)
- Free text feedback.

D. Sample Layout

[背景圖：森林]

故事文本：

你站在濃霧中，聽見狼嚎...

> 玩家輸入區：

[掛起火把並前進]

[ 語音] [ TTS]

按鈕：

[Regenerate] [Edit AI Text]